RESEARCH SYNTHESIS / VERSION 1.0

Trading Edge

Information, risk, constraints, and the discipline of getting paid

A research-backed operating method for LLMs that create, improve, and test trading strategies.

THE OPERATING QUESTION

What do we know, provide, or bear more efficiently than the marginal alternative - and can we capture enough value to survive the costs, competition, and bad states?

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Includes economic foundations, worked mathematics, ten falsifiable research blueprints, validation templates, and an annotated primary-source register.

1 / The realization and its implications

The realization in the post

Johann Markov's screenshot proposes two origins of edge: superior information or access to favorable bets, and accepting risks other people would rather transfer. The screenshot was inspected directly. An original post permalink was not independently located, so this study interprets the supplied text; it does not attribute additional claims or a complete trading method to its author.

The powerful shift is from looking for a pattern to explaining a transaction. An indicator describes observations. An economic edge explains why a feasible action offers value at its price, why someone else accepts the other side, and why the advantage survives competition. The counterparty may have different information, but may also have a rational reason to pay for insurance, liquidity, immediacy, funding, or relief from a mandate. An expected transfer of cash can coexist with an improvement in both parties' welfare.

The post is a productive research lens rather than an exhaustive theorem. Its two categories overlap: a market maker can both know more and warehouse inventory. Access, execution, contract design, subsidies, and allocation can change attainable returns without fitting neatly into a mutually exclusive taxonomy. A strategy's economics need a payoff model; labeling it does not estimate it.

The central synthesis

A defensible edge is a repeatable, participant-specific advantage in the conversion of information, services, or risk-bearing capacity into feasible net economic value. This is an operational definition proposed by this study. It deliberately includes three comparisons: against the current price, against competitors' capabilities, and against the investor's actual alternative use of capital.

That definition changes the questions an LLM should ask. What precisely is the traded payoff? Which observations arrive before the decision? What does the price already reveal? What changes when our order fills? Who needs the transaction? What state makes the position painful? How much of the apparent profit pays for that exposure? Which rival strategy can provide the same service more cheaply? What would make us stop believing the claim?

The output of this research is therefore a reusable method for creating and improving hypotheses, with executable specifications and decisive tests. It cannot supply missing observations or turn a mechanism into demonstrated profit. Its purpose is to make the model substantially harder to fool and much better at deciding where to investigate.

Three refinements that matter most

Information is valuable relative to a price and a decision. A 70% chance of an event is a bad purchase at a price requiring 80%. Accurate forecasts on events nobody can trade add little trading value. The relevant question is whether new evidence changes a feasible decision enough to pay for collecting it, interpreting it, and acting before the price moves. Costs and endogenous price impact limit how much of an informational advantage can be collected. [I01, I02, I06, B01]

Risk is valuable to the seller only at an adequate premium. Variance alone does not entitle its bearer to compensation. A loss that coincides with a customer's most expensive state can command a premium; an independent avoidable gamble may not. A warehouse must be able to survive the period when collateral requirements rise and liquidity vanishes. Terminal convergence does not pay a margin call today. [R1, R2, R3]

A credible story and a credible test do different jobs. The story suggests when an effect should appear, its capacity, and its failure conditions. Measurement determines whether a policy adds value at executable prices. A convincing explanation can accompany a false discovery, while a reproducible empirical effect can exist before its mechanism is fully understood. Both uncertainty and alternative explanations must remain visible. [V1, V10, V12]

How the disciplines connect

Discipline	What it contributes	What an LLM must not infer
Bayesian decision theory	Evidence updates distributions; actions depend on utility and prices	More confident language is a stronger likelihood ratio
Information economics	Information is costly; prices partly aggregate it	Public information is automatically useless or automatically mispriced
Game theory	Competitors infer, react, and choose under constraints	A profitable historical opponent remains exploitable forever
Microstructure	Fills, spreads, queues, impact, and latency select realized opportunities	Every displayed quote can be captured at size
Gambling and betting	Odds make break-even, bankroll, and payout asymmetry explicit	High hit rate, Kelly, or stake progression creates edge
Prediction markets	Forecasts interact with preferences, wealth, contract rules, and market design	A quoted price equals truth or similarly named contracts are identical
Asset pricing and insurance	Bad-state exposure and hedging demand can justify expected returns	Every risk premium is skill, or more variance means more compensation
Psychology	Attention, repetition, overconfidence, and disposition can affect behavior	A documented bias necessarily affects the marginal price enough to trade
Statistical science	Separate discovery, selection, confirmation, and monitoring	A low p-value after a large hidden search certifies the winner
Engineering and control	Preserve causal timing, capital, reproducibility, and correct execution	More complexity or faster polling necessarily improves economic value

A concrete illustration of the change in thinking

Suppose a model estimates a binary event probability of 0.56. A contract paying one unit costs 0.52, and deterministic all-in costs are 0.015 per contract. Its modeled expected net payoff is 0.025. That is a hypothesis about a payoff, not a discovered profit stream.

Now suppose passive orders fill mainly during states where the conditional event probability is 0.50. The filled contract has expected payoff -0.035. The original forecast can be reasonably calibrated across all opportunities and still fail where the strategy receives fills. Improving a language model's broad prediction accuracy may do less than measuring this selection effect and changing the execution policy. All numbers here are illustrative, and the conditional probability must be estimated prospectively rather than assumed. [I03; original calculation]

A second hypothetical strategy wins one unit 99% of the time and loses 150 units otherwise. Its expected payoff is -0.51. Calling it risk warehousing does not change that arithmetic. Even a positive

premium would require joint tail, liquidity, and capital analysis before it could be considered suitable.

Designing intelligence that is economically useful

The strongest agentic desk is not the one that thinks hardest about every observation. It is the one that spends the next unit of analysis where it can change the decision before the opportunity disappears. This study proposes testing an adaptive research policy: use a cheap calibrated model first; request additional evidence only when estimated decision improvement exceeds its money and delay costs; stop at a defined decision deadline. Repeated news and multiple agents drawing on one source must count as one evidence lineage. This is an untested design hypothesis, grounded in decision value and attention theory, not a claim of achieved alpha. [I06, I07, I11]

Likewise, the best improvement to a strategy may be a subtraction: remove an unhelpful filter, stop trading opportunities whose costs exceed their forecast advantage, reduce overlapping tail exposure, or eliminate timing leakage. Another improvement may be a different use of the same signal: a slower execution schedule, a more suitable payoff contract, or a smaller position that retains favorable marginal economics. These are candidates to compare against the frozen baseline, not universal prescriptions. [S3, S4]

Scope and evidence policy

The following chapters combine academic primary papers, author manuscripts, institutional research records, and official contract documentation across the requested disciplines. The source register identifies versions and whether the work was examined through available body text, abstract, or metadata. It does not claim exhaustive access to every book, interview, article, or source on the internet. Historical findings are used to generate and constrain research, not to assert that the same returns remain available today.

The mathematical examples are hypothetical and the research blueprints are untested. No market backtest, broker execution, or live deployment was performed for this study. Formula checks and prompt-behavior evaluations validate the deliverable's internal reasoning and operating instructions; they are not strategy-performance evidence.

Read the foundations for the economic model, the mechanism blueprints for creation and improvement, and the validation protocol for deciding what evidence has actually been earned. The separate master prompt provides the operational instructions in a form that can be pasted into another LLM. The installed skill adds selective access to the full reference material.

2 / Economic and mathematical foundations

1. Interpret the two-part insight

Johann Markov's post groups edge into information arbitrage and risk warehousing. The useful insight is to look behind a profitable pattern and ask what scarce contribution earns the return: improving a decision before competitors do, or supplying exposure, immediacy, patience, or capital that others value.

Grossman-Stiglitz explains why costly information need not be completely revealed by equilibrium prices: complete revelation would undermine its production incentive. That leaves room for information rents, but those rents may merely cover acquisition costs [I01]. Risk warehousing has another economic justification: counterparties can rationally pay to reduce an exposure even when everyone agrees about its distribution [R1, R5].

The categories overlap. A liquidity supplier can forecast order flow, bear inventory risk, and earn a service payment simultaneously. Access, market design, mandates, financing, subsidies, execution, and portfolio complementarity also matter [I12, R7, B09, B10]. Some are causes of gross return; others determine whether a particular trader captures it. The post is therefore a powerful organizing lens, not an exhaustive theorem. “Variance” also needs refinement: volatility alone does not establish compensation.

2. Separate expected profit, alpha, and utility

Define three distinct claims before evaluating a strategy:

• **Positive expected net profit:** the conditional mean of the complete cash payoff, after relevant costs, is positive. Compare funded investments with their financing and opportunity costs.

• **Alpha:** performance exceeds a specified feasible benchmark or modeled risk exposures. Alpha depends on that benchmark; it does not establish immunity to omitted risks.

• **Incremental utility:** adding the trade improves the owner's objective given existing assets, liabilities, funding, and constraints. This can differ from standalone expected return.

A hedge can have negative expected excess return and improve portfolio utility. A leveraged market exposure can earn positive expected return without forecasting alpha. Better execution can preserve an existing gross edge without creating new information [R1, R10].

Hypothetical example: a payoff earns 1 with probability 0.99 and loses 150 otherwise. Its mean is −0.51. Neither the 99% win rate nor a sequence of small profits establishes favorable risk compensation. Conversely, positive expectation does not guarantee attractive utility when losses coincide with the owner's most vulnerable states.

3. Identify the marginal counterparty

The relevant competitor is whoever determines the executable price at the intended size and time, not an average participant. Identify the objective behind the opposing order: a hedge, liquidity need, mandate, inventory adjustment, different evidence, or superior knowledge. An apparent seller of cheap risk may be the better-informed party [B06].

Milgrom-Stokey's no-trade result makes this question rigorous under its assumptions: private information does not automatically create mutually acceptable speculation starting from an efficient

allocation with appropriate belief and common-knowledge conditions [I04]. Real opportunities require explaining which conditions differ. Aumann similarly makes persistent disagreement a question about evidence, priors, models, and common knowledge - not proof that our estimate is superior [I05].

Execution itself selects information. **Derivation:** for a single-fill order with quantity n,

$$EV(a) = P(F \mid \mathcal{I}, a)\, E[n(V_h - P_{fill}) - C \mid \mathcal{I}, a, F] - C_{attempt}.$$

Here F denotes a fill; C includes applicable exit and holding costs. Expand partial-fill outcomes when necessary. Do not multiply by fill probability again if an empirical payoff series already includes zero-payoff unfilled attempts. Resting orders can receive inventory precisely when their valuation becomes unfavorable: evaluate conditional markouts and cancellation delays [I03].

4. Understand risk compensation

Asset pricing expresses a payoff's value as

$$P = E[mX], \qquad E[R] - R_f = -\frac{\mathrm{Cov}(m, R)}{E[m]}, \quad R_f = 1/E[m].$$

Here m is a stochastic discount factor, X a terminal payoff, and R a gross return [R1]. Payoffs delivered when marginal capital is valuable can command high prices and low expected returns. Losses in those states may require compensation. Independent variance alone does not imply a premium.

The unwanted burden may be a crash, unhedgeable jump, uncertain convergence time, margin call, borrow recall, or illiquidity. Option demand can affect prices through residual hedging risks [R5]. Funding losses and widening spreads can reinforce one another [R3]. A terminally correct convergence thesis can fail through interim liquidation [R2].

Operational deduction: define liquid equity W, required collateral M, and reserve B along each stress path; require W−M−B to satisfy the desk's funding constraint. Do not substitute initial margin or a historical stop distance for this calculation. Examine simultaneous spread widening, losses, haircut changes, and inaccessible collateral. Distinct strategies can warehouse the same crisis exposure despite low ordinary correlations [R4].

5. Read prices and contract semantics correctly

A price is a payoff valuation, not automatically a physical probability. For a digital contract paying one unit if A occurs,

$$P_A = E[m\mathbf{1}_A] = P(A)E[m \mid A].$$

With a positive discount factor and bond price B=E[m], P_A/B defines the corresponding risk-neutral probability; it need not equal P(A). Prediction-market probability interpretations also depend on preferences, wealth, beliefs, hedging motives, and restrictions [B04]. Treat prices as an informative benchmark, not an oracle.

For two outcomes offered at decimal odds 1.91, raw reciprocal odds sum to 1.04712. Proportional normalization gives 0.50/0.50, but that assumption does not recover objective probabilities. Different margins and customer demand can distort the mapping [B05, B06].

The traded event must match its contractual definition: measurement source, cutoff, inclusivity, revisions, voids, and settlement delay [B12]. A correct forecast of the headline can be a wrong forecast of the payout.

Hypothetical example: a one-dollar binary payoff priced at 0.52 with all-in cost 0.015 and physical probability 0.56 has expected net profit 0.025. If the filled subset instead has probability 0.50, its expectation is −0.035. Price advantage must survive both contract interpretation and execution selection.

6. Update beliefs without counting evidence twice

Derivation: posterior odds obey

$$O(H \mid \mathcal{I}, E_{1:k}) = O(H \mid \mathcal{I}) \prod_{j=1}^{k} \frac{P(E_j \mid H, \mathcal{I}, E_{<j})}{P(E_j \mid \neg H, \mathcal{I}, E_{<j})}.$$

The conditional terms matter. With prior probability 0.50, one observation carrying likelihood ratio 2 moves the posterior to 2/3. Three copies of that observation still supply one underlying observation. Treating them as independent would falsely produce 8/9. Source lineage, first-publication times, revisions, and shared inputs belong in the evidence model.

Tetlock's research connects stale news to subsequent reversal in its studied sample [I11]. That motivates novelty tracking; it does not imply that repeated information is always irrelevant. Changed context can change an old fact's significance.

Likewise, multiple indicators calculated from one price history and multiple LLMs using one article are not independent experiments. A transformation may improve a bounded model's representation without adding external information [I06]. Estimate likelihoods and calibration from defensible data or models; fluent confidence is not a likelihood ratio. Update on prices too when they reveal additional information, while avoiding circular use of the same signal.

7. Connect information to attainable decisions

More accurate prediction can be economically useless when the action stays unchanged, prices already incorporate it, or delivery is late. **Derivation:** the expected sample-information value for experiment Z is

$$EVSI = E_Z[\max_a E[U(a,\theta) \mid \mathcal{I}, Z]] - \max_a E[U(a,\theta) \mid \mathcal{I}] - c_Z.$$

Include abstention among feasible actions. This formula assumes the stated posterior and decision model; empirical implementations must estimate attainable value rather than assume an oracle. Blackwell's comparison of experiments provides the decision-theoretic foundation [I06]. Sims supplies a framework for scarce attention [I07].

Hypothetical example: choose between buying one binary contract for 0.55 or passing. Prior success probability is 0.50. An experiment produces posterior 0.80 or 0.20 with equal probability while the quote remains available. Buying only after the favorable result yields expected profit 0.5×0.25=0.125; a 0.03 experiment cost leaves 0.095. If the price updates before execution, this value changes or disappears.

Public economic links and competing demands on attention motivate research into delayed interpretation [I09, I10]. Speed must be measured against executable opportunity lifetime: end-to-end latency and order size define a capture surface, not a universal “faster is better” rule [I02, I12].

8. Derive the information-growth identity

Assume mutually exclusive exhaustive states, complete frictionless state bets, normalized strictly positive prices q_i summing to one, true probabilities p_i, and a fully invested nonnegative allocation b_i summing to one. Require b_i>0 wherever p_i>0. A unit invested with allocation b pays b_i/q_i in state i. **Derivation:**

$$E_p[\log(W_1/W_0)] = \sum_i p_i \log(b_i/q_i) = D_{KL}(p\|q) - D_{KL}(p\|b).$$

Information relative to market odds creates potential log-growth value; allocation error consumes it. This sharpens Kelly's information connection [B01]. It is a special-model identity, not a universal alpha equation. Incomplete contracts, costs, credit, leverage, and changing prices alter the feasible calculation. Repetition requires suitable assumptions before interpreting the expectation as a long-run growth rate.

Hypothetical example: p=(0.60,0.40), q=(0.50,0.50), b=(0.80,0.20). Market divergence is 0.020136 nats; allocation error is 0.104650; expected log growth is −0.084514. Using b=p instead gives +0.020136 in this ideal model. Knowing the more likely outcome is insufficient when confidence and stakes are wrong.

Proper scoring rules evaluate forecast distributions [B03]. Keep forecast score and economic results separate: improvements outside the traded subset need not improve realized utility.

9. Allocate under uncertainty and survival constraints

For a binary bet with net win odds d−1, probability p, and wealth fraction f,

$$G(f,p) = p\log(1+(d-1)f) + (1-p)\log(1-f), \qquad f^* = \frac{pd-1}{d-1}.$$

For long-only betting, constrain the solution to the feasible set, including zero. At p=0.56 and d=1.91, expected profit per dollar staked is 0.0696 and ideal f*=0.07648. These are hypothetical known-model results, not recommended stakes [B01, B02].

Parameter uncertainty alone does not algebraically imply half-Kelly. Because G is linear in p, averaging this one-period objective over a posterior uses its mean: E_p[G(f,p)]=G(f,E[p]). Shrinkage can change that mean. Drawdown preferences, model ambiguity, correlation, and funding constraints introduce additional reasons to change allocation [V9].

For a long-only ambiguity interval $p \in [0.52, 0.60]$, the worst-case probability is 0.52; at odds 1.91 it implies no favorable bet. This robustness protects only against the specified uncertainty set [B08]. Risk-constrained Kelly supplies modeled drawdown restrictions, whose assumptions and initial-wealth floor must not be confused with universal protection or a rolling maximum-drawdown guarantee [B07]. Optimize joint wealth, not isolated per-trade fractions.

10. Recognize strict arbitrage

In a specified frictional model, strict arbitrage requires a feasible self-financing position with nonpositive initial cost, no adverse future payoff across the relevant states, and a strictly positive benefit somewhere with positive probability. Every contingent cash requirement must be financeable. A positive mean, likely convergence, or attractive relative valuation does not meet that definition.

Hypothetical example: exactly complementary outcomes have executable decimal odds 2.10 and 2.05. Their reciprocal sum is 0.963995. Allocating 493.9759 and 506.0241 from a 1,000 stake equalizes the payout at 1,037.3494, a 3.73494% ideal return before costs. This conclusion fails if legs cannot be executed together, quantities differ, outcomes overlap, void rules differ, or a venue does not pay.

Real research must establish payoff identity, simultaneous executable capacity, settlement and counterparty assumptions, fees, and financing. Market structure can create mechanical opportunities while intense competition makes them inaccessible to a particular desk [I12]. A scarce-access rent and a universally available arbitrage are different claims.

11. Transfer ideas across disciplines

Discipline	Useful transfer	Boundary
Gambling	Separate favorable odds from bankroll allocation [B02].	House take and limits can consume apparent advantage.
Prediction markets	Model exact event semantics and benchmark probabilities [B04, B12].	Price aggregates incentives as well as beliefs.
Poker/game theory	Distinguish robust play from opponent exploitation [B11].	Market rules, competitors, and external utility evolve.
Insurance	Identify the liability someone pays to transfer [R5].	Rare losses and dependence determine capital needs.
Psychology	Translate bias into observable demand and price effects [V6, V7].	Individual bias alone does not establish market alpha.
Statistical learning	Evaluate prospective forecasts and search-adjusted evidence [V1, V8].	Predictive fit does not establish capture or causality.

These connections suggest a productive research question: where can this desk supply information, inventory, patience, or implementation at lower all-in cost than the marginal alternative? They do not authorize assuming that it already can.

12. Distinguish a persuasive mechanism from evidence

An economic mechanism guides falsification, transfer, and failure diagnosis. It cannot replace measurement. Research comparing theoretical and mechanically generated predictors cautions against using an appealing explanation as a certificate of durability [V12]. Conversely, a reproducible unexplained signal can remain a candidate with greater uncertainty.

Maintain separate judgments about mechanism, forecast quality, executable capture, and portfolio suitability. Register all explored variants; preserve chronology and dependence; compare feasible benchmarks; and log decisions before outcomes [V1, V2, V8]. DSR and PBO diagnose particular selection problems, not the probability of future profitability [V3, V4].

Competition and capital can change the opportunity after discovery [V5, V11]. The skill's achievement is therefore to produce a falsifiable candidate and an honest evidence status. Its strongest output may be a precisely identified bottleneck - or a demonstrated absence of edge at the available price.

3 / Ten research blueprints

How to use these blueprints

Every blueprint below is an **untested research hypothesis**, an original synthesis for this skill, and a starting specification. None establishes current profitability, historical performance, or priority of discovery. Source identifiers refer to `sources.md`. Their findings motivate questions; they do not validate the proposed policies.

Select the blueprint that matches an observable mechanism and available data. Replace symbolic thresholds with values selected inside training periods under a declared mandate. Freeze the candidate family, benchmark, principal outcome and rejection rule before confirmation. Follow `validation.md` for chronological testing, dependent observations, trial accounting and prospective logging. Retain abstention as an action. A falsifier defeats the specified claim; insufficient power leaves it unresolved.

Mechanism and discipline map

Blueprint	Economic object	Connected discipline
1	Underprocessed new evidence	Bayesian inference; attention
2	Selection into realized trades	Auction theory; calibration
3	Compulsory demand	Institutional finance; causal inference
4	State-dependent marginal risk	Insurance; portfolio utility
5	Occupied collateral	Queueing intuition; capital allocation
6	Exact state payoffs	Logic; linear programming
7	Cost of risk removal	Insurance pricing; optimization
8	Value of another observation	Decision theory; bounded rationality
9	Value decay under impact	Microstructure; optimal control
10	Strategic response	Game theory; adaptive markets

1. Novel information through economic links

Anchors: [I01], [I07], [I09], [I11]. **Hypothesis:** genuinely new information can reach a customer's price before a less-followed supplier's price because interpreting the economic connection consumes attention. Syndicated repetitions should not create additional Bayesian evidence.

As-of data: first-public timestamps, source lineage, document revisions, previously disclosed customer exposure, contemporaneous quotes, news load and sector returns. Freeze each relationship to its public disclosure date.

Minimal policy: map a new event into a signed exposure estimate; trade a bounded position only when the estimated residual payoff clears executable costs and a predeclared uncertainty hurdle. Use

a fixed holding horizon or information invalidation; skip ambiguous exposure and stale news.

Rival: common industry shocks, ordinary momentum or retrospectively chosen relationships explain the effect. **Experiment/baseline:** compare a market-and-sector model, a simple customer-return model and the semantic model on identical chronological events; add matched unlinked firms and irrelevant-event placebos.

Falsifier: no incremental net value, or gains require later-known links. **Implementation risk:** incorrect entity resolution, stale contracts, duplicate stories, and arrival latency can convert apparent insight into already-priced information.

2. Calibration conditional on execution

Anchors: [I03], [I08], [B03], [V8]. **Hypothesis:** forecast quality can deteriorate precisely when a resting order receives a fill. Counterparties select when to trade, so unconditional calibration over all timestamps need not describe inventory actually acquired.

As-of data: submitted price, observed quotes, volatility, order age, acknowledgements, partial fills, cancellations and subsequent markouts. Log every attempt, including zeros and rejects.

Minimal policy: estimate both fill probability and payoff distribution conditional on each execution outcome. Choose passive, marketable or skip by expected net utility, including nonfill opportunity cost; cancel under a predefined stale-quote rule.

Rival: a change in signal quality or market state explains poor fills, rather than adverse selection. **Experiment/baseline:** hold forecasts and position limits fixed; compare immediate crossing, naive passive execution and conditional execution. Report scores within execution-state bins and outcomes over all attempts.

Falsifier: conditional routing adds no held-out net benefit or relies on impossible queue fills. **Implementation risk:** cancellation requests are not acknowledgements; partial inventory remains exposed. Avoid subtracting adverse selection twice when conditional payoff estimates already include it.

3. Forced flow versus fundamental news

Anchors: [R6], [R9], [R11], [V10]. **Hypothesis:** observable compulsory transactions can create temporary pressure, while information-driven transactions can justify lasting repricing. A profitable reversal policy must distinguish them before entry.

As-of data: announced index rules and changes, public auction information, dated holdings, actual publication lags, quotes, earnings/news timestamps and contemporaneous risk exposures. A falling price alone does not establish forced selling.

Minimal policy: estimate pressure from a documented obligation; enter only when estimated mechanical demand is large relative to available liquidity and competing news does not invalidate the valuation anchor. Define a maximum holding period and loss/capital limits.

Rival: sellers anticipate deteriorating fundamentals; the supposed pressure measure proxies for bad news. **Experiment/baseline:** compare ordinary reversal, pressure-only and pressure-plus-news policies on matched events with sector, liquidity and momentum controls. Seek externally determined rule changes for stronger identification.

Falsifier: the proposed pressure interaction adds no executable reversal or vanishes after point-in-time controls. **Implementation risk:** incomplete holdings, front-running competitors and long

convergence delays can make an economically plausible effect inaccessible.

4. Risk complementarity across the desk

Anchors: [R1], [R3], [R4], [B07]. **Hypothesis:** a compensated exposure can be more valuable to one desk than another because their existing liabilities and crisis exposures differ. The comparative advantage may reside in the portfolio rather than superior prediction.

As-of data: synchronized sleeve returns, current positions, liabilities, collateral terms, cash reserves and scenario sensitivities; include unavailable marks and funding calls.

Minimal policy: allocate only if a candidate improves a declared portfolio objective after costs while satisfying joint loss and collateral constraints. Evaluate a candidate's incremental exposure, not its standalone Sharpe. Allow zero allocation despite positive expected profit.

Rival: apparent complementarity reflects short samples, stale prices or calm-period correlation. **Experiment/baseline:** compare equal-capital allocations, matched-risk allocations and the proposed scenario-aware policy using chronological and resampled crisis scenarios. Include a cash allocation alternative.

Falsifier: improvement disappears when common funding shocks or plausible tail dependence are included. **Implementation risk:** multiple brokers may not recognize economic offsets for margin; correlated withdrawals can destroy presumed liquidity. This blueprint improves suitability and capital use; it does not create new gross alpha.

5. Capital occupation and convergence time

Anchors: [R2], [R3], [R9], [B01]. **Hypothesis:** ranking opportunities by headline spread misallocates scarce capital when funding duration and interim collateral needs differ. Faster release can be valuable, but only if replacement opportunities actually exist.

As-of data: executable spreads, contractual maturities, resolution histories including failures, borrow/funding terms, haircut schedules and the contemporaneous opportunity set.

Minimal policy: estimate joint distributions of net payoff, convergence time and peak capital demand. Select a feasible portfolio maximizing the stated horizon objective against a maturity-matched cash alternative; prohibit assumed unlimited rollover.

Rival: shorter trades merely hide larger tail or reinvestment risk. **Experiment/baseline:** compare gross-spread ranking, simple expected-return ranking and duration-aware selection at identical liquid capital and stress limits. Replay the opportunity stream without future knowledge.

Falsifier: duration estimates provide no incremental utility or gains depend on unrealistic reuse of released cash. **Implementation risk:** settlement delay, borrow recall and margin expansion can extend the occupation window. Expected profit per capital-day is a screening statistic, not a complete objective or reliable annualized return.

6. Exact contracts and payoff constraints

Anchors: [B04], [B09], [B12], [R2]. **Hypothesis:** related contract prices can be inconsistent once their exact state payoffs are represented correctly; alternatively, precise semantics may explain an apparent inconsistency completely.

As-of data: contract wording and revisions, authoritative outcome source, boundary inclusivity, observation clock, void rules, executable depth, fees, collateral and settlement terms.

Minimal policy: create a complete relevant state set and payoff matrix A. Represent each permitted long/short leg with its own executable cost and quantity bound. Search by linear programming for a portfolio x whose net state payoffs meet a specified floor after funding, or minimize cost for an exactly matched target payoff. Claim arbitrage only if every permitted outcome and financing path supports it; otherwise classify a risky spread.

Rival: mismatched definitions, unavailable shorting or stale quotes generate the apparent gap. **Experiment/baseline:** compare headline matching with audited semantic matching and replay full executable bundles.

Falsifier: no feasible positive surplus survives complete states and costs. **Implementation risk:** atomic execution may be unavailable; unfilled legs, disputed resolution and cross-venue collateral cannot be removed by an algebraic proof.

7. Hedging versus simply trading less

Anchors: [R1], [R5], [R8], [B08]. **Hypothesis:** a hedge can improve portfolio utility when it removes a costly shared exposure more efficiently than reducing all profitable positions. Insurance may also consume the entire premium.

As-of data: executable hedge surfaces, full portfolio scenarios, physical payoff forecasts, risk-neutral prices, financing and margin rules. Keep horizons and units consistent when comparing implied and forecast variance.

Minimal policy: choose among unhedged smaller exposure, bounded exposure plus a specified hedge, and cash. Include hedge premium, renewal, basis, exercise and stressed liquidation in the joint objective.

Rival: the attractive equity curve results solely from lower exposure or an omitted insurance cost. **Experiment/baseline:** compare fixed hedging, state-aware hedging and simple deleveraging under equal capital and comparable tail constraints; freeze the hedge menu before testing.

Falsifier: the hedge offers no incremental utility beyond lower size after realistic prices and stress dependence. **Implementation risk:** protection can expire before the loss, fail on basis, or be unavailable during stress. A cheaper option is not necessarily cheaper protection against the relevant state.

8. Adaptive research expenditure

Anchors: [I06], [I07], [B03], [V1]. **Hypothesis:** scarce research time is best spent where another observation can improve a feasible decision before its opportunity expires. More analysis can reduce value through cost, delay and correlated evidence.

As-of data: sequential source arrivals, acquisition/computation costs, model versions, forecast snapshots, retrieval failures, action changes and quote changes during analysis.

Minimal policy: start with a calibrated inexpensive model. Estimate incremental decision value for each available research action; purchase the most valuable observation only while its conservative estimated benefit exceeds money and delay cost. Stop at the deadline or retain the baseline action.

Rival: apparent benefits arise from hindsight selecting which cases deserved deeper analysis. **Experiment/baseline:** train routing on earlier events; compare shallow, uniformly deep, simple threshold routing and learned routing at matched budgets. Evaluate the entire sequential policy.

Falsifier: routing adds no prospective net utility or fails after charging latency. **Implementation risk:** LLM fluency can masquerade as information gain; ten agents repeating one source supply one evidence lineage. Record routing variants in the trial ledger.

9. Latency and capacity together

Anchors: [I02], [I12], [R10], [V8]. **Hypothesis:** a signal's usefulness is a surface over delay, size and execution mode. Faster infrastructure matters only when additional captured value exceeds its cost; larger orders may erase the signal through impact.

As-of data: exchange/event, vendor-arrival, decision, submission, acknowledgement and fill timestamps; observed depth, actual participation, fees and benchmark prices. Audit clock alignment.

Minimal policy: estimate net value V(delay, quantity, mode); choose the feasible combination subject to uncertainty, deadline and capital limits. Integrate depth across the complete order rather than pricing every unit at the best quote.

Rival: measured latency decay is timestamp error, revised data or execution selection. **Experiment/baseline:** replay the same frozen signals at deliberate delays and size grids; compare simple marketable execution and a slower low-cost system. Confirm key regions with prospective shadow records.

Falsifier: positive value exists only below attainable latency, at unavailable size, or before total infrastructure costs. **Implementation risk:** exchange priority, queue cancellation and transient depth defeat naive replay; infrastructure speed cannot rescue an incorrect payoff forecast.

10. Opponent response and opportunity decay

Anchors: [B06], [B11], [I04], [V5], [V11]. **Hypothesis:** an exploitable behavior can weaken when counterparties learn, capital enters or venue rules change. A stable fallback and bounded exploitation may outperform permanent commitment to one observed weakness.

As-of data: dated rule/fee changes, spreads, markouts, participant proxies, forecasts, capacity and strategy versions. Treat counterparty identity and intent as uncertain unless directly documented.

Minimal policy: retain a feasible benchmark policy; permit a bounded deviation when prospective evidence supports incremental value. Review on a declared schedule or valid sequential detector; choose continue, reduce, pause or retire without unconstrained retuning.

Rival: ordinary sampling variation or exposure changes mimic strategic adaptation. **Experiment/baseline:** compare frozen exploit, frozen benchmark and prespecified adaptive policy; replay every review decision chronologically and examine forecast deterioration separately from capture deterioration.

Falsifier: adaptation fails to improve net utility or false alarms dominate avoided losses. **Implementation risk:** reacting to every loss mines noise, and switching itself incurs cost. Poker equilibrium analogies do not guarantee financial profitability; counterparties may rationally accept negative cash expectation for hedging utility.

Example: improving an indicator strategy

Suppose an existing rule buys a closed-bar breakout when volume exceeds a threshold. Preserve its original entry clock, universe, size, stops and exits as the frozen baseline. The indicator describes an event; it does not yet identify the payer or mechanism.

Propose two competing explanations: new public information is still propagating, or temporary compulsory demand is exhausting itself. Use blueprint 1 to construct a first-publication novelty variable and blueprint 3 to identify independently documented mechanical-flow events. Do not label high volume itself as either mechanism.

Run a small, registered comparison: baseline; baseline plus novelty; baseline excluding documented flow events; and the combined rule. Measure all eligible opportunities, including skipped trades. Match capital and exposure so a filter cannot win merely by trading less. Use chronological event splits and charge actual observation-to-entry delay; a close-confirmed signal cannot assume an earlier closing fill.

Apply blueprint 2 if gross prediction survives but passive fills lose money. Compare the same forecasts under immediate crossing and fill-conditioned execution before adding indicators. Apply blueprint 7 only if portfolio tail exposure remains the binding problem.

Report where improvement occurs: forecasting, selection, capture, cost or portfolio utility. If the novelty variable adds no prospective value, remove it. If all benefit is reduced market exposure, describe that accurately. Respect any designated timing authority: research filters and execution changes must not silently replace the approved entry engine. No experiment here has been run; the output is a reproducible improvement plan with explicit alternatives.

4 / Validation and autonomous-desk handoff

1. Decision artifacts

Complete the following compact Edge Contract before selecting a final candidate. Write `UNKNOWN` where evidence is missing; distinguish a measured value from an assumption.

```
contract_id / version / research_owner:
mode: CREATE | IMPROVE | AUDIT
decision: what capital or policy decision will this evidence change?
estimand: net expectancy | benchmark excess | portfolio utility | forecast loss
instrument / exact payoff / venue / horizon / eligible universe:
participant segment / objective / constraint:
proposed transfer: information error | paid service/risk | unresolved pattern
our measurable advantage / executable access:
observable state / action / exit / abstention:
information availability / decision deadline / execution policy:
persistence mechanism / capacity limit / adverse state:
strongest rival explanation:
primary observable implication / negative control:
falsifier / missing evidence / cheapest decisive experiment:
benchmark / economic hurdle / error tolerance / risk mandate:
current evidence stage / permitted next stage:
```

Maintain three additional artifacts:

Artifact	Required fields
Experiment card	Hypothesis; one conceptual change; estimand; eligible opportunities; benchmark; training/evaluation dates; outcome horizon; planned contrasts; costs; dependence unit; inference method; power/precision assumptions; promotion/failure rule
Append-only trial ledger	Trial and parent IDs; creation time; researcher/agent; rationale; code/config/data hashes; universe and feature versions; parameter values; splits; label definition; execution assumptions; metrics viewed; outcome; rejected variants; holdout-access history
Decision record	Evidence stage; completed versus unrun tests; estimates and uncertainty; execution feasibility; stress results; strongest contradiction; selected action; unresolved conditions; next review trigger; approver if required by mandate

Count agent-generated variants, discarded filters, alternative labels, and changed cost assumptions as research choices. Unknown earlier searches make selection-adjusted inference incomplete. Do not silently record only successful experiments. [V1, V3]

2. CREATE and IMPROVE workflows

For **CREATE**, first evaluate whether each proposed mechanism has observable implications and feasible access. Select a small research set with genuinely different mechanisms, then specify a cheap falsification experiment for each. For example, an information-delay thesis predicts weaker returns when entry is delayed; a forced-flow thesis predicts variation with the observable constraint. Check a rival explanation that produces the same pattern. A chart association alone cannot establish the proposed causal transfer. [V10]

Build a minimal policy, freeze its candidate family, and conduct chronological development and validation. Fit preprocessing, feature selection, thresholds, and calibration inside training periods. Reserve a final period that has not informed choices. If the policy adapts, replay the entire learning schedule using only information then available. Test the selected policy prospectively before moving

its evidence stage.

For **IMPROVE**, freeze the existing policy and reproduce its outputs first. Attribute the shortfall to forecast quality, opportunity selection, entry/exit, costs, fill quality, exposure, or portfolio interaction. Choose the binding problem and one conceptual intervention; include simplification as a candidate. Compare policies on the same external opportunity stream, calendar, capital convention, and feasible execution assumptions. Include abstentions and missed opportunities rather than comparing only each policy's successful fills.

Use paired differences where appropriate, preserving shared event dependence. Match exposure or explicitly attribute changed beta, volatility, tail loss, and capital usage. Test interactions after evaluating individual changes. A filter that reduces turnover may improve net results without improving forecasting; name the improvement correctly. Additional leverage changes returns without establishing additional edge.

For both modes, select controls that isolate the claim: cash, passive exposure, matched-risk exposure, delayed signal, removed feature, same execution without the new model, or event-time randomization within defensible blocks. Unrestricted shuffling can erase dependence and create an irrelevant null. Keep the benchmark and principal metric fixed through confirmation. Reusing holdout results to choose revisions makes that sample development data. [V1, V2]

3. Availability, fills, and dependence

Construct a decision-level evidence table:

```
opportunity_id, instrument_id, event_id, event_time,
source_publication_time, vendor_receipt_time, revision_id,
feature_available_time, decision_time, information_cutoff,
model_version, forecast, action_or_abstention,
quote_time, bid, ask, executable_size,
order_time, order_id, acknowledgment_time,
fill_time, fill_price, fill_size, fees,
exit_or_settlement_time, realized_payoff, rejection_reason
```

Use immutable instrument identifiers and historical membership; retain delistings, expired instruments, revisions, and missing-data outcomes. Verify that every feature was available by the decision cutoff. Final daily bars, subsequently confirmed pivots, revised releases, and retrospectively named regimes require explicit availability treatment. Trade simulation must begin at an achievable time after the decision.

For passive orders, estimate fill probability and payoff conditional on filling; a fill can select adverse states. Preserve no-fill branches and opportunity cost. For multiple legs, simulate partial execution, hedge/unwind costs, and collateral use. Use executable sides and size rather than treating the midprice as attainable. Report results under plausible spread, impact, financing, and latency stress; missing execution evidence lowers the evidence stage.

Record the true dependence unit. Multiple trades in one event, overlapping holding intervals, securities exposed to one announcement, and repeated labels of the same future move do not constitute independent observations. Purge training labels whose outcome intervals overlap test outcomes. Where future training observations are retained in a resampling design, use justified separation for information carried from the test interval. Select gaps from actual information and holding horizons.

Resample blocks or event clusters while keeping competing policies aligned. Examine plausible block-length choices and cross-sectional clustering. Fit any learned transformations within the relevant training fold. Dependence correction does not eliminate regime change or repair leakage.

4. Statistical tools and limits

Choose methods for the question, not for a desired verdict:

Tool	Appropriate interpretation	Incorrect interpretation
Reality Check	Tests whether the searched family contains predictive superiority over the benchmark, accounting for the recorded search	Proves the winning economic mechanism or repairs unrecorded searches [V1]
SPA	Studentized comparative test less sensitive to poor alternatives; global rejection indicates at least one superior candidate	Certifies whichever individual strategy is subsequently selected [V2]
DSR	Selection- and non-normality-aware Sharpe diagnostic using trial and distribution information	Posterior probability of live profitability; substitute for valid costs or dependence assumptions [V3]
PBO/CSCV	Estimates instability of a specified selection process: in-sample winners ranking below the candidate median out-of-sample	Probability of a monetary loss; replacement for chronological deployment replay [V4]

Predeclare one principal decision procedure and use other methods as diagnostics. Do not average incompatible statistics into a fictitious probability. Explain conflicting results. Candidate-specific promotion after a global test requires suitable follow-up inference. Distinguish insufficient evidence from evidence of no economically useful effect. Repeated interim looks need a valid sequential design or explicit multiplicity adjustment.

Score prospective forecasts against timestamp-matched base-rate and market forecasts. Evaluate calibration and discrimination separately from decision economics: a calibrated base rate may offer no useful discrimination, while improved scoring may occur outside tradable states. [V8]

5. Quantitative precision example

Hypothetical illustration, not a backtest or allocation recommendation. A binary bet pays decimal odds $d = 1.90$; deterministic additional cost is $c = 0.02$ per unit stake. Expected net payoff is $1.90p - 1.02$; break-even probability is $1.02 / 1.90 \approx 0.53684$. An estimated $p = 0.56$ implies estimated net expectancy 0.044 per unit stake. It does not establish that the true probability exceeds break-even.

Suppose the experiment prospectively chooses approximate 95% two-sided probability precision of ±0.02. Under independent, identically distributed Bernoulli outcomes, a conservative normal-approximation calculation gives

$n \approx 1.96^2 \times 0.25 / 0.02^2 = 2{,}401$.

If that sample actually produced an estimate near 0.56, its approximate lower endpoint near 0.54 would narrowly exceed the original break-even probability. Raising cost to 0.04 moves break-even to $1.04 / 1.90 \approx 0.54737$, eliminating that confidence-bound conclusion.

This is a precision illustration, not a power calculation, a measured confidence interval, or a universal trade requirement. For planning power against a specific alternative, also choose the acceptable miss probability and calculate the corresponding sample requirement. Use suitable exact/score intervals when appropriate. Correlated outcomes, changing probabilities, selected bets, and repeated strategy search invalidate the simple independent calculation; cluster-aware precision and selection correction

may require substantially more data. Variable payoffs require inference on net payoff or utility rather than win probability alone.

6. Keep, modify, pause, or reject

Before confirmation, define economic hurdle h, false-promotion tolerance, principal test family, capital-loss limit, collateral buffer, acceptable model uncertainty, and stress scenarios. Derive them from the objective and cost of a wrong decision. Do not impose universal Sharpe, win-rate, trade-count, or calendar-duration gates.

• **Keep:** the existing policy remains within mandate, implementation is valid, and review evidence does not trigger a prespecified change. Keeping is not a claim of certainty.

• **Promote:** valid candidate-specific evidence supports net benefit above h, execution is feasible, relevant stress fits capital constraints, and replay reproduces the policy. Promotion is limited to the next authorized stage.

• **Modify:** attribution identifies a repairable bottleneck and a specific experiment can distinguish improvement from a rival explanation. Create a new version; preserve prior results.

• **Pause:** data, reconciliation, settlement, collateral, or execution validity fails, or a prescribed risk boundary is crossed. Investigate the operational cause before interpreting P&L.

• **Reject/retire:** a preregistered falsifier is observed, credible costs consume the opportunity, the policy violates mandate, or adequate evidence rules out the required economic benefit.

• **Inconclusive:** uncertainty remains too large. State what additional information would change the decision and whether obtaining it is worthwhile.

A risk warehouse need not profit during every stress; its adverse states must be understood and capitalized. Historical maximum drawdown is not a future loss bound. Avoid forced retraining after ordinary losses: adaptation can become hidden repeated search. [V5, V11]

7. Forward monitoring and deterministic integration

Write forecasts and intents before outcomes. Monitor forecast score/calibration, eligible-opportunity coverage, fill probability, fill-conditioned markouts, realized costs, latency, exposure, collateral, and net attribution. Separate forecast deterioration from capture deterioration. Track the mechanism's observable driver alongside P&L; publication and competition can alter opportunity and correlation. [V8, V11]

Specify review cadence, sequential evidence rules, retraining windows, rollback versions, and independent operational kill conditions. Keep hard execution deadlines and risk checks deterministic. Persist the input snapshot, model/config versions, intent identifier, broker acknowledgments, and reconciled state. Make submission idempotent, reconcile after restart, and reject stale inputs or unknown position state. Test partial fills, duplicate events, outages, and restored state against the approved policy.

LLMs may produce research candidates and structured explanations; policy changes require the same versioned replay and acceptance process as other changes. End the handoff with actual evidence and deployment status, reproducible commands, unresolved dependencies, and the single next experiment most likely to alter the decision.

5 / Source register

Checked October 23, 2026, America/New_York. Read this register to trace economic, statistical, and market-mechanics claims to their source and inspect their limits. The research was broad and targeted; it was not an exhaustive search of the internet or complete reading of every cited work.

The user's Johann Markov screenshot was viewed directly. Its original post permalink was not independently located. Treat the screenshot as the supplied thesis, not independent verification of its author's broader views.

There are **50 distinct source entries**: 49 research/book/manuscript entries and one official documentation group containing two Kalshi pages. There are 51 stable labels because **V9 aliases B08**. Related manuscripts R10 and S4 are separately identified versions/projects, not independent corroboration. Alternate URLs and publication versions do not add to the count. The screenshot is not counted. Access descriptions distinguish inspected text from abstracts and indexed records; none implies every page was parsed.

Published results establish the claims stated below under their assumptions. Skill workflows, examples, diagnostic formulas, and proposed research programs are engineering synthesis or elementary derivations unless explicitly attributed. A historical finding is not evidence that a present strategy is profitable.

Information and game theory

I01 - Grossman and Stiglitz (1980), On the Impossibility of Informationally Efficient Markets

Sanford J. Grossman; Joseph E. Stiglitz. *American Economic Review* 70(3), 393-408. Institutional record.

Access: institutional record and available primary-paper material inspected. **Supports:** costly information creates a tension between fully revealing prices and incentives to acquire information. Residual informativeness can compensate information production. **Limit:** this equilibrium mechanism does not establish any particular signal's excess return; gross compensation may only repay acquisition and implementation costs. Require an explicitly costed information advantage.

I02 - Kyle (1985), Continuous Auctions and Insider Trading

Albert S. Kyle. *Econometrica* 53(6), 1315-1335. Publisher.

Access: indexed publisher abstract; PDF retrieval failed. **Supports:** informed strategic trading, noise demand, and competitive pricing jointly determine information revelation and price impact. **Limit:** the stylized market is not a modern execution simulator, and its theoretical informed trader is not permission to use unlawful information. A signal's value depends on attainable size, timing, and endogenous impact.

I03 - Glosten and Milgrom (1985), Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders

Lawrence R. Glosten; Paul R. Milgrom. *Journal of Financial Economics* 14(1), 71-100. Author institution; DOI90044-3).

Access: primary institutional research record inspected. **Supports:** adverse selection can produce a positive spread despite a risk-neutral, zero-expected-profit specialist. **Limit:** quoted spread is not realizable profit; a fill changes the relevant conditioning information. Estimate fill-conditioned value and post-fill markouts. Do not deduct adverse selection twice if it already enters the payoff distribution.

I04 - Milgrom and Stokey (1982), Information, Trade and Common Knowledge

Paul Milgrom; Nancy Stokey. *Journal of Economic Theory* 26(1), 17-27. Author PDF.

Access: author PDF and indexed publication record inspected. **Supports:** specified Pareto-efficiency, belief, and common-knowledge conditions prevent mutually acceptable speculative trade generated solely by private information. **Limit:** liquidity needs, constraints, incomplete markets, and different priors violate relevant assumptions. The theorem does not forbid real-world trading or require irrationality for profit; identify the proposed mechanism's broken assumption.

I05 - Aumann (1976), Agreeing to Disagree

Robert J. Aumann. *Annals of Statistics* 4(6), 1236-1239. Author institution; DOI.

Access: author institutional record and indexed original-paper material inspected. **Supports:** common-prior Bayesian agents cannot retain different event posteriors when those posteriors are common knowledge. **Limit:** market prices are not transparent beliefs, and theorem assumptions often fail. Diagnose disagreements through evidence, priors, likelihoods, payoff definitions, and constraints; repeated LLM agreement using the same evidence is not independent confirmation.

I06 - Blackwell (1953), Equivalent Comparisons of Experiments

David Blackwell. *Annals of Mathematical Statistics* 24(2), 265-272. Publisher.

Access: publisher/indexed record inspected; publisher rendering limited full-text access. **Supports:** under its conditions, comparison of experiments by garbling corresponds to attainable decision payoff across decision problems. **Limit:** this is a partial ordering, not a claim that every extra datum helps a particular trade. Information costs and bounded-model representation quality must be analyzed separately from new external evidence.

I07 - Sims (2003), Implications of Rational Inattention

Christopher A. Sims. *Journal of Monetary Economics* 50(3), 665-690. Author institution; DOI00029-1).

Access: institutional primary publication record inspected. **Supports:** finite information-processing capacity between observations and actions can generate behavior resembling adjustment frictions. **Limit:** a general attention model does not prove an exploitable anomaly. Specify the relevant information bottleneck, economic stakes, and action change; neglect can be rational when the attainable dollar opportunity is smaller than research and execution costs.

I08 - Milgrom and Weber (1982), A Theory of Auctions and Competitive Bidding

Paul R. Milgrom; Robert J. Weber. *Econometrica* 50(5), 1089-1122. University-hosted PDF; DOI.

Access: primary-paper PDF and publication record inspected. **Supports:** affiliated signals and interdependent values make rival behavior informative; conditional valuation addresses winner's-curse effects. **Limit:** not every market order is a common-value auction. Ask what obtaining a fill, winning an allocation, or selecting the highest apparent opportunity reveals, and test performance conditional on that selection process.

I09 - Cohen and Frazzini (2008), Economic Links and Predictable Returns

Lauren Cohen; Andrea Frazzini. *Journal of Finance* 63(4), 1977-2011. Author institution; Coauthor research page.

Access: author/institutional research records inspected. **Supports:** historical customer-supplier return predictability is consistent with limited attention to public economic relationships. **Limit:** published historical results do not establish current after-cost returns. Links must be dated to their public availability, and tests must control for own momentum, industries, liquidity, and shared exposures before crediting semantic propagation.

I10 - DellaVigna and Pollet (2009), Investor Inattention and Friday Earnings Announcements

Stefano DellaVigna; Joshua M. Pollet. *Journal of Finance* 64(2), 709-749. Author institution; Author manuscript.

Access: primary institutional research record inspected. **Supports:** the sample's Friday announcements have weaker immediate response, stronger delayed response, and lower volume, supporting an attention interpretation. **Limit:** announcement timing is selected and the effect may decay. Test attention proxies and alternative explanations; a calendar label alone neither identifies the mechanism nor establishes a presently executable trade.

I11 - Tetlock (2011), All the News That's Fit to Reprint: Do Investors React to Stale Information?

Paul C. Tetlock. *Review of Financial Studies* 24(5), 1481-1512. Author institution; DOI.

Access: primary institutional research record inspected. **Supports:** textual similarity identifies stale stories whose associated price moves predict reversals, particularly with greater individual-investor activity. **Limit:** similarity is an imperfect novelty measure, and repeated information can acquire new relevance. Preserve provenance, first-publication times, and revisions; syndicated copies must not become independent Bayesian likelihood updates.

I12 - Budish, Cramton, and Shim (2015), The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response

Eric Budish; Peter Cramton; John Shim. *Quarterly Journal of Economics* 130(4), 1547-1621. Author research page.

Access: author research page and primary results inspected. **Supports:** continuous serial processing can create speed rents even with symmetrically public information; batch auctions change the competitive margin. **Limit:** market-design findings do not establish that a generic bot can capture those rents. Compare opportunity lifetime with measured observation, decision, submission, acknowledgement, and fill delays.

Risk, constraints, and implementation

R1 - Cochrane, Asset Pricing, author-hosted sample chapters (copyright 2001)

John H. Cochrane. Author page; Sample chapters. A revised edition appeared in 2005; this retrieved file was **not verified as that edition**.

Access: sample-chapter text, copyright page, and pricing equations inspected. **Supports:** expected excess return relates to covariance with the stochastic discount factor, not raw variance alone. **Limit:** the identity becomes an empirical risk explanation only with a specified pricing kernel and market assumptions. Name the economically bad state and whether the strategy loses when capital is most valuable.

R2 - Shleifer and Vishny (1997), The Limits of Arbitrage

Andrei Shleifer; Robert W. Vishny. *Journal of Finance* 52(1), 35-55. Published PDF; Author record.

Access: published opening sections inspected. **Supports:** specialized arbitrage capital can shrink after losses as investors withdraw precisely when price discrepancies widen. **Limit:** persistent spreads can also reflect omitted risk or contract differences. Specify interim mark-to-market loss, collateral, investor withdrawals, and convergence uncertainty; a correct terminal view does not prevent forced liquidation before convergence.

R3 - Brunnermeier and Pedersen (2009), Market Liquidity and Funding Liquidity

Markus K. Brunnermeier; Lasse Heje Pedersen. *Review of Financial Studies* 22, 2201-2238. Author page; Paper.

Access: abstract, model, and margin sections inspected. **Supports:** capital losses, funding constraints, and market illiquidity can reinforce one another, creating shared liquidity exposures. **Limit:** liquidity spirals are conditional outcomes rather than inevitable responses to volatility. Stress joint price loss, spreads, haircuts, margin, and liquidation; initial broker margin is not an adequate capital requirement for strategy design.

R4 - He, Kelly, and Manela (2017), Intermediary Asset Pricing: New Evidence from Many Asset Classes

Zhiguo He; Bryan Kelly; Asaf Manela. *Journal of Financial Economics* 126, 1-35. Published PDF.

Access: published abstract and introductory empirical framework inspected. **Supports:** innovations in primary-dealer capital ratios explain cross-sectional expected returns across the studied asset classes. **Limit:** explanatory evidence is not a reliable live timing rule. Account for release lags, composition changes, and uncertainty, and test whether apparently diversified strategies all depend on healthy intermediary balance sheets.

R5 - Gârleanu, Pedersen, and Poteshman (2009), Demand-Based Option Pricing

Nicolae Gârleanu; Lasse Heje Pedersen; Allen M. Poteshman. *Review of Financial Studies* 22(10), 4259-4299. January 2006 manuscript; Publication record.

Access: 2006 manuscript theorem and residual-hedging-risk sections; 2009 publication verified. **Supports:** demand affects prices through unhedgeable variance and covariance of residual hedging risk. **Limit:** volume and open interest do not reveal net customer demand or dealer inventory. Separate compensation for bearing residual option risk from unusually favorable pricing relative to that same risk.

R6 - Coval and Stafford (2007), Asset Fire Sales (and Purchases) in Equity Markets

Joshua Coval; Erik Stafford. *Journal of Financial Economics* 86, 479-512. Published PDF.

Access: published abstract and discussion inspected. **Supports:** flow-driven mutual-fund transactions create price pressure in overlapping holdings in the 1980-2004 sample. **Limit:** disclosure lags and endogenous flows complicate replication; older manuscript samples differ. Identify observable obligations to trade using decision-time information rather than calling every decline a forced sale or using subsequently disclosed holdings.

R7 - Frazzini and Pedersen (2014), Betting Against Beta

Andrea Frazzini; Lasse Heje Pedersen. *Journal of Financial Economics* 111, 1-25. Published PDF.

Access: introduction and proposition summaries inspected. **Supports:** leverage constraints can elevate high-beta asset prices; the studied low-beta/high-beta construction earns returns but suffers when funding constraints tighten. **Limit:** beta neutrality is neither funding neutrality nor tail neutrality. Financing, short access, estimation error, and mandate-specific constraints determine whether the proposed investor can capture the documented mechanism.

R8 - Bollerslev, Tauchen, and Zhou (2009), Expected Stock Returns and Variance Risk Premia

Tim Bollerslev; George Tauchen; Hao Zhou. *Review of Financial Studies* 22(11), 4463-4492. Publisher abstract; DOI.

Access: publisher abstract and bibliographic information; full paper not read in this pass. **Supports:** implied-versus-realized variation relates to subsequent aggregate returns, especially intermediate quarterly horizons in the study. **Limit:** this is not evidence for 0DTE selling. Distinguish risk-neutral variance expectations, physical forecasts, and later realized variance; their differences contain different economic and statistical components.

R9 - Duffie (2010), Presidential Address: Asset Price Dynamics with Slow-Moving Capital

Darrell Duffie. *Journal of Finance* 65(4), 1237-1267. Author-hosted published PDF.

Access: abstract and opening framework inspected. **Supports:** search and capital-raising frictions can create sharp price responses followed by extended reversals as capital reaches opportunities slowly. **Limit:** this does not justify fading every information shock. Specify the source of the imbalance, mechanism of capital arrival, convergence-time uncertainty, and the funding required to survive the intervening path.

R10 - Frazzini, Israel, and Moskowitz (August 2018 working paper), Trading Costs

Andrea Frazzini; Ronen Israel; Tobias J. Moskowitz. Institution-hosted manuscript; Author submission.

Access: abstract and execution/measurement sections inspected. **Supports:** institutional executions permit estimates of impact and implementation shortfall, including missed-trade costs. **Limit:** the studied executions were almost always completed, and stock-lending fees are excluded. These costs are not universal for small stocks, retail options, or fast signals. Estimate size, urgency, adverse selection, nonfills, and borrow separately.

R11 - Gabaix and Koijen (May 12, 2022 manuscript), In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis

Xavier Gabaix; Ralph S. J. Koijen. Retrieved manuscript; NBER record.

Access: manuscript abstract, identification, and robustness passages inspected. **Supports:** relatively inelastic institutional demand can amplify market-level flow effects, including through portfolio mandates. **Limit:** estimated aggregate multipliers are not universal intraday or stock-specific impact constants. Distinguish temporary liquidity shocks from durable allocations, and treat inferred demand elasticities and participant identities as uncertain estimates.

Betting, forecasting, and allocation

B01 - Kelly (1956), A New Interpretation of Information Rate

J. L. Kelly Jr. *Bell System Technical Journal* 35(4), 917-926. Publisher; Article reproduced with permission.

Access: article text inspected. **Supports:** connects information about gamble outcomes with attainable long-run log wealth growth at offered odds. **Limit:** optimal allocation depends on the payoff and probability model; the ideal result does not remove costs, estimation error, limited horizons, or constraints. Correct directional forecasts can still lose

wealth through miscalibration or excessive stakes.

B02 - Thorp (1997; revised May 29, 1998), The Kelly Criterion in Blackjack, Sports Betting, and the Stock Market

Edward O. Thorp. Conference paper. Primary text and chapter links.

Access: abstract and structure read; chapter PDFs retrieved but yielded no extracted text. **Supports:** separates finding favorable bets from deciding their size and discusses growth-optimal allocation across gambling and securities. **Limit:** Thorp's historical success does not validate a new system. Kelly, martingale staking, or equity smoothing cannot by itself turn a negative-expectation payoff into positive-expectation alpha.

B03 - Gneiting and Raftery (2007), Strictly Proper Scoring Rules, Prediction, and Estimation

Tilmann Gneiting; Adrian E. Raftery. *Journal of the American Statistical Association* 102(477), 359-378. Author PDF.

Access: original paper text inspected. **Supports:** strictly proper scores reward honest probabilistic forecasts in expectation and connect forecasting with entropy, calibration, and sharpness. **Limit:** scoring improvement does not imply executable profit, especially on the selected trade subset. Maintain separate forecast and net-policy evaluations; financial chronology and dependence require suitable validation rather than indiscriminate random folds.

B04 - Wolfers and Zitzewitz (2006), Interpreting Prediction Market Prices as Probabilities

Justin Wolfers; Eric Zitzewitz. FRBSF Working Paper 2006-11 / NBER 12200. Primary paper.

Access: paper sections on utility, wealth, and equilibrium inspected. **Supports:** prices can equal mean beliefs under specific conditions; correlated wealth and beliefs produce wealth-weighted aggregation. **Limit:** average belief is not objective truth, and hedging, utility, restrictions, and costs affect demand. Treat price as a strong benchmark with explicit assumptions, never automatic proof for or against model disagreement.

B05 - Snowberg and Wolfers (2010), Explaining the Favorite-Long Shot Bias: Is It Risk-Love or Misperceptions?

Erik Snowberg; Justin Wolfers. *Journal of Political Economy* 118(4), 723-746. Primary working-paper repository.

Access: primary abstract read; NBER PDF indexed but direct retrieval denied. **Supports:** compound-bet choices help distinguish risk preference from probability misperception, favoring the latter in the studied horse-racing data. **Limit:** this does not establish universal longshot overpricing or profitable favorites after house costs. Test mechanism-specific predictions and actual net value in the target market.

B06 - Levitt (2004), Why Are Gambling Markets Organised So Differently from Financial Markets?

Steven D. Levitt. *Economic Journal* 114(495), 223-246. Author/university PDF.

Access: original article text inspected. **Supports:** bookmakers in the studied setting deliberately take outcome exposure and exploit bettor biases instead of only balancing wagers. **Limit:** dated sportsbook evidence is not a universal description of current exchanges. A counterparty accepting inventory may also be better informed; observed imbalances alone do not establish uninformed flow or a hedging need.

B07 - Busseti, Ryu, and Boyd (2016), Risk-Constrained Kelly Gambling

Enzo Busseti; Ernest K. Ryu; Stephen Boyd. *Journal of Investing* 25(3), 118-134. Author PDF.

Access: model and drawdown framework inspected. **Supports:** a tractable conservative bound can constrain drawdown probability while optimizing log growth. **Limit:** the core repeated-return model assumes IID draws; the initial-wealth floor is not a rolling peak-to-trough maximum-drawdown guarantee. Unmodeled dependence, tails, collateral, or changing regimes invalidate any unsupported claim of real-market protection.

B08 - Sun and Boyd (2018 preprint; revised 2021), Distributional Robust Kelly Gambling: Optimal Strategy under Uncertainty in the Long-Run

Qingyun Sun; Stephen Boyd. Primary preprint, version 3 dated June 10, 2021. Also cited as **V9**.

Access: author preprint abstract and version information inspected. **Supports:** optimize worst-case expected log growth over a specified set of distributions; useful finite-state cases are tractable. **Limit:** protection extends only to the specified ambiguity set and wealth model. Too narrow a set creates false reassurance; broad sets can favor cash. An unexplained fractional-Kelly haircut is not this result.

B09 - Hanson (2007; author draft January 2002), Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation

Robin Hanson. *Journal of Prediction Markets* 1(1), 3-15. Author draft; Published paper.

Access: author draft inspected; publication version verified. **Supports:** market scoring rules combine forecasting incentives with group information aggregation and useful conditional consistency properties. **Limit:** information subsidies are paid by someone, and marginal quotes change with size. Classify rebates and sponsored liquidity separately from persistent alpha, and integrate the execution price curve rather than extrapolating the first quote.

B10 - Cover (1991), Universal Portfolios

Thomas M. Cover. *Mathematical Finance* 1(1), 1-29. Original article; Inspected author version dated October 23, 1996.

Access: author version inspected; 1991 publication verified. **Supports:** a nonanticipating allocation asymptotically matches the exponential growth rate of the best constant-rebalanced portfolio in hindsight under the specified model. **Limit:** relative regret is not guaranteed positive absolute return, finite-horizon superiority, or costless implementation. Distinguish diversification and allocation improvements from new predictive information and evaluate feasible benchmarks.

B11 - Brown and Sandholm (2019), Superhuman AI for Multiplayer Poker

Noam Brown; Tuomas Sandholm. *Science* 365(6456), 885-890. Author paper.

Access: original paper's game-theory discussion inspected. **Supports:** strong multiplayer poker performance and the distinction between equilibrium robustness and opponent exploitation; special two-player zero-sum guarantees do not transfer automatically. **Limit:** financial markets have changing participants, uncertain payoff models, and external utility. Poker success is not evidence of LLM market alpha; model counter-adaptation and the disappearance of exploitable behavior.

B12 - Kalshi (March 17, 2026), Market Rules; Market Outcomes

Official documentation group: Market Rules; Market Outcomes. Two primary pages, counted as one documentation entry.

Access: both pages read. **Supports:** individual contracts specify resolution criteria and information sources; determination need not coincide with trading closure. **Limit:** general help pages are not a particular contract's terms. Verify cutoff, inclusivity, revisions, exceptions, settlement, collateral, and current fees for the exact instrument. Similar headlines do not establish economically equivalent payoffs or arbitrage.

Validation, behavior, and adaptation

V1 - White (2000), A Reality Check for Data Snooping

Halbert White. *Econometrica* 68(5), 1097-1126. Original paper.

Access: original paper inspected. **Supports:** judging a selected winner requires accounting for the candidate search, rather than testing it as the only attempted model. **Limit:** inference only reflects the recorded candidate universe and the procedure's dependence assumptions. Hidden failures and later revisions remain material; predictive superiority neither identifies a causal mechanism nor proves economically sufficient executable returns.

V2 - Hansen (2005), A Test for Superior Predictive Ability

Peter Reinhard Hansen. *Journal of Business & Economic Statistics* 23(4), 365-380. Author research page; Primary submission.

Access: author/SSRN abstracts inspected; publisher open unavailable. **Supports:** SPA uses studentization and a sample-dependent null to reduce sensitivity to irrelevant poor alternatives. **Limit:** rejecting the global null establishes superiority somewhere in the candidate family, not automatic certification of its selected winner. Preserve joint dependence during resampling and prespecify the principal test instead of selecting favorable diagnostics.

V3 - Bailey and López de Prado (2014), The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality

David H. Bailey; Marcos López de Prado. Author version July 31, 2014. Paper.

Access: author paper inspected. **Supports:** assesses Sharpe inflation using selection, track-record length, non-normality, and the distribution of trials. **Limit:** it is a model-based significance diagnostic, not a posterior probability of live profit. Unknown search histories, leakage, dependency misspecification, stale executions, and regime change remain unresolved. Document the effective trial count rather than treating correlated variants as independent.

V4 - Bailey, Borwein, López de Prado, and Zhu (February 27, 2015 author version), The Probability of Backtest Overfitting

David H. Bailey; Jonathan M. Borwein; Marcos López de Prado; Qiji Jim Zhu. Author paper.

Access: paper and rank-based definition inspected. **Supports:** combinatorially symmetric cross-validation estimates selection instability: how often the in-sample winner ranks below the out-of-sample median. **Limit:** PBO is not the probability of losing money; one minus PBO is not a profit probability. Combinatorial partitions do not replace chronological evaluation reproducing every live refit, selection decision, and execution restriction.

V5 - Lo (2004), The Adaptive Markets Hypothesis: Market Efficiency from an Evolutionary Perspective

Andrew W. Lo. *Journal of Portfolio Management*, 30th anniversary issue. Author paper.

Access: primary paper inspected. **Supports:** competition, capital, institutions, and conditions can change efficiency and strategy opportunity sets. **Limit:** this developing framework is not a quantitative law supplying an optimal regime variable or adaptation speed. Test those choices prospectively; automatic retuning after losses may merely repackage data mining as adaptation, and changed conditions need not explain every drawdown.

V6 - Odean (1998), Are Investors Reluctant to Realize Their Losses?

Terrance Odean. *Journal of Finance* 53(5), 1775-1798. Author paper.

Access: primary paper inspected. **Supports:** records for 10,000 accounts show a disposition toward realizing gains over losses that common alternative explanations do not adequately explain in the sample. **Limit:** individual bias does not establish a tradable price effect. Show that affected holders influence the marginal price, can be identified using available data, and create a persistent net opportunity.

V7 - Barber and Odean (2000), Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors

Brad M. Barber; Terrance Odean. *Journal of Finance* 55(2), 773-806. Author paper.

Access: paper and abstract inspected. **Supports:** highly active households in the studied 1991-1996 brokerage sample had worse net performance, consistent with overconfidence and trading costs. **Limit:** the association is not experimental proof of every trade's motivation, and historical retail costs are not current universal costs. Activity and confident language are not evidence of incremental net utility.

V8 - Dawid (1984), Present Position and Potential Developments: Some Personal Views - Statistical Theory: The Prequential Approach

A. Philip Dawid. *JRSS Series A* 147(2), 278-290. Publisher; Paper.

Access: publisher summary inspected. **Supports:** assess inference through sequential forecasts issued before future observations; this motivates timestamped prospective prediction records. **Limit:** calibration alone is insufficient: a base-rate model can be calibrated but uninformative, and an informative model may be untradeable at prevailing prices. Compare market-relative forecast quality separately from execution and decision value.

V9 - Alias of B08: Sun and Boyd, Distributional Robust Kelly Gambling

Use **B08** for the exact source, 2018/2021 version information, access description, supported result, and limitations. This stable cross-reference is preserved for validation materials but **does not count as a second source**. Distributional ambiguity must be represented explicitly; it is not interchangeable with a posterior probability interval, a generic haircut, or a guarantee against unmodeled losses.

V10 - Dawid (2020), Decision-theoretic Foundations for Statistical Causality

A. Philip Dawid. Author preprint.

Access: author paper record inspected. **Supports:** transferring observational evidence to intervention decisions requires justified relationships between observational and decision regimes. **Limit:** profitable prediction need not identify every causal parameter, but attributing the opportunity to a specific participant or mechanism requires stronger evidence than fit. Trading at scale can itself alter the relevant regime through market impact and strategic response.

V11 - McLean and Pontiff (2016), Does Academic Research Destroy Stock Return Predictability?

R. David McLean; Jeffrey Pontiff. *Journal of Finance* 71(1), 5-32; first online 2015. Publisher.

Access: publisher abstract inspected. **Supports:** returns to 97 predictors declined out of sample and further after publication, consistent partly with publication-informed trading; predictor correlations also increased. **Limit:** sample averages are not a universal haircut or proof every public strategy fails. Diagnose current durability and crowding prospectively rather than converting historical decay percentages into fixed rules.

V12 - Chen, Lopez-Lira, and Zimmermann (2022 working paper), Does Peer-Reviewed Research Help Predict Stock Returns?

Andrew Y. Chen; Alejandro Lopez-Lira; Tom Zimmermann. Retrieved working-paper record. No claim that this is the latest journal version.

Access: author working-paper record inspected. **Supports:** the studied data-mined accounting predictors can have post-sample performance comparable to published predictors, challenging theory labels as automatic durability evidence. **Limit:** this is a defined comparison, not a theorem that mechanisms are useless. Use mechanisms for falsification and transfer diagnostics, while requiring measurement rather than substituting persuasive economic stories for evidence.

Additional foundations

S1 - Sharpe (1991), The Arithmetic of Active Management

William F. Sharpe. *Financial Analysts Journal* 47(1), 7-9. Author text.

Access: primary author text inspected. **Supports:** for the defined market and accounting conventions, aggregate active management matches passive holdings before costs and underperforms after higher costs. **Limit:** aggregate arithmetic neither excludes individual skill nor identifies exploitable counterparties. State the market, benchmark, investor group, and costs; a positive active return can still be a negative contribution relative to opportunity cost.

S2 - Pedersen (2018), Sharpening the Arithmetic of Active Management

Lasse Heje Pedersen. *Financial Analysts Journal* 74(1), 21-36. Publisher/research page.

Access: primary publication abstract inspected. **Supports:** changing securities supply and real implementation complicate static active-versus-passive arithmetic; passive portfolios also need trading. **Limit:** qualifying an accounting benchmark does not prove a selected active strategy earns net alpha. Define feasible replication, issuance/reconstitution trading, and costs before using a frictionless passive benchmark or claiming active

management necessarily adds value.

S3 - Gârleanu and Pedersen (2013), Dynamic Trading with Predictable Returns and Transaction Costs

Nicolae Gârleanu; Lasse Heje Pedersen. *Journal of Finance* 68(6), 2309-2340; NBER working paper 15205 (2009). NBER record; March 2012 author manuscript.

Access: primary abstract verified and March 2012 author PDF retrieved; NBER direct open returned 403. **Supports:** optimal trading balances predictive-return decay with trading costs and future target positions. **Limit:** a model-derived policy depends on its signal, cost, and risk assumptions. Faster turnover need not improve capture when additional impact overwhelms the signal retained.

S4 - Frazzini, Israel, and Moskowitz (2012 working paper), Trading Costs of Asset Pricing Anomalies

Andrea Frazzini; Ronen Israel; Tobias J. Moskowitz. Author research page.

Access: primary research abstract inspected. **Supports:** institutional execution data can change assessments of anomaly costs, implementation, and capacity; portfolio construction matters for attainable returns. **Limit:** institutional execution evidence is not a universal fee or impact schedule for another desk. Related authorship and research lineage with R10 do not constitute independent confirmation. Re-estimate costs for the intended assets, scale, and urgency.

6 / Deliverable checks and use

What was checked

The skill metadata and structure passed the skill-creator validator. All source identifiers used by the references resolve to the source register. The register contains 51 labels for 50 distinct entries; V9 is an explicit alias of B08. Worked numerical examples were independently recalculated. The PDF equations were rendered as mathematical images and the page layouts visually inspected.

Two fresh agents received the skill and supplied hypothetical tasks, without being given the intended answers or prior diagnoses. They produced user-facing solutions to four application cases. The review below checks those outputs against the supplied arithmetic, payoff definitions, and research requirements. This is a small qualitative behavior check; it does not measure superiority over a baseline prompt, prove generalization to every LLM, or validate any market strategy.

Case	Observed behavior	Review result
Accurate forecast at bad odds, plus passive-fill selection	Derived -0.06 per contract for p=.60 at price 0.65 plus cost 0.01; derived -.035 per passive fill and -.014 per attempt under fill-only costs; distinguished overall accuracy from conditional probability	Arithmetic and diagnosis matched supplied assumptions
Apparent cross-venue weather arbitrage	Enumerated mismatched station outcomes; found the state where both contracts lose and pair P&L is -.90; designed a semantic payoff checker	Rejected the false risk-free claim and gave a constructive implementation path
Sharpe 3.1 and 98% win rate selected from 2,000 backtests	Kept the result at unverified research status; identified contaminated dates, unknown search history, midpoint-fill optimism and downside tails; specified an execution audit and shadow workflow	Did not manufacture a corrected significance estimate or live-readiness claim
New customer-to-supplier earnings strategy	Specified a long-only, delayed public-information hypothesis with as-of relationships, explicit features, decision clocks, entry/exit rules, baselines, costs and falsifiers	Produced a reproducible research candidate; labeled numerical settings assumptions and returns unmeasured

What to invoke

Use the saved skill named engineer-trading-edge, displayed as Trading Edge Lab. A practical starting request is:

"Use $engineer-trading-edge to audit this strategy. Identify its economic source of edge and the main loss of value, then execute accessible research and produce the smallest useful improvement with an explicit experiment and rejection rule."

For a different LLM, paste Trading_Edge_Master_Prompt.txt and append the strategy, available data, instruments, horizon, objective, and constraints. The master prompt is self-contained; this study supplies deeper references and research blueprints. Its purpose is to create and test better hypotheses, not to substitute prose for evidence.